

Online Neural Space Time Memory for Dynamic Novel View Synthesis

Anonymous WACV Algorithms Track submission

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A. Additional Implementation Details

In this section, we provide comprehensive details regarding our training curriculum, optimization hyperparameters, baseline configurations, and the evaluation protocol for our generalized Novel View Synthesis experiments.

A.1. Two-Stage Training Curriculum

All stateful models are trained sequentially on short video clips consisting of $T = 4$ timesteps. At each timestep, the network processes 2 sparse input views to synthesize 1 target view. To optimize both the memory persistence and the free-viewpoint rendering flexibility, we employ a two-stage training curriculum, as follows.

Stage 1: Memory Bootstrapping (Pretraining). The objective of this initial phase is to explicitly encourage the network to build and rely upon its memory module. For the first frame of the sequence ($t = 0$), the input consists of one frontal view (`cam00`) and one back-facing view (`cam08`), providing broad geometric context. For subsequent frames ($t > 0$), the inputs transition to two random views. Crucially, the target supervision is a target view that is explicitly distinct from the $t > 0$ inputs. Note these target viewpoints remain *static* across the sequence. This rigid setup boosts memorizing the initial context to faithfully inpaint the targets at later timesteps.

Stage 2: Free-Viewpoint Adaptation (Finetuning). Once the memory module is bootstrapped, we adapt the network to handle completely unconstrained streaming scenarios. In this stage, the inputs at $t = 0$ are two random views, and the inputs at $t > 0$ are two random *other* views. Furthermore, the target viewpoints are no longer static; they are completely random, time-varying views that are strictly disjoint from the 4 input views. Ultimately, this allows for continuous free-viewpoint rendering from time-varying target views (e.g., a 360-degree spin around the subject) while continuously querying the generalized memory.

A.2. Training Hyperparameters

We implement our framework using JAX. Models are trained with a total batch size of 128 for 500,000 itera-

tions during the pretraining stage, followed by an additional 25,000 iterations during the finetuning stage. We employ a cosine learning rate decay schedule with a linear warmup. During pretraining, the learning rate is initialized at 1×10^{-6} , warms up over 4,000 steps to a peak of 4×10^{-4} , remains constant for 100,000 steps, and then decays down to a final value of 1×10^{-6} . During the finetuning stage, we retain the same schedule parameters but reduce the peak learning rate to 4×10^{-5} warming up over 4,000 steps and then remaining constant. Finally, for the image reconstruction objective across all models and stages, the perceptual loss weight is fixed at $\lambda_{\text{lips}} = 0.5$.

A.3. Baseline Adaptations and Coordinate Alignments

To ensure a fair comparison, all baselines evaluated in our main paper are trained on the exact same two-training curriculum. However, due to architectural differences, camera coordinate systems must be aligned differently:

LVSM [2]: We utilize the exact architecture described in the original LVSM paper. But for fair comparison, we use the same learning rate schedule as above and the same L2 and LPIPS loss. Because LVSM is a stateless, per-frame model, we construct its training batches by extracting only the final frame ($t = 3$) from our 4-timestep video clips, ensuring it trains on identical data distributions. For coordinate alignment, we relativize the camera coordinate system to each individual target camera over time (i.e., the specific target camera for that frame is set as the origin with an identity rotation).

LaCT [4] and NSTM (Ours): Stateful models must maintain coherent camera coordinate system across time and adapt to time-varying target cameras. Therefore, for LaCT and NSTM, we relativize all cameras in the sequence to the *first* target camera ($t = 0$), ensuring a stable, global coordinate space for the fast-weights to operate within.

Token-Mem [3]: As explained in our main paper, *Token-Mem* is a custom novel view synthesis model whose memory mechanism is based on CUT3R [3]. As a stateful method, it also requires a temporally consistent coordinate frame, and we use the same coordinate system as described

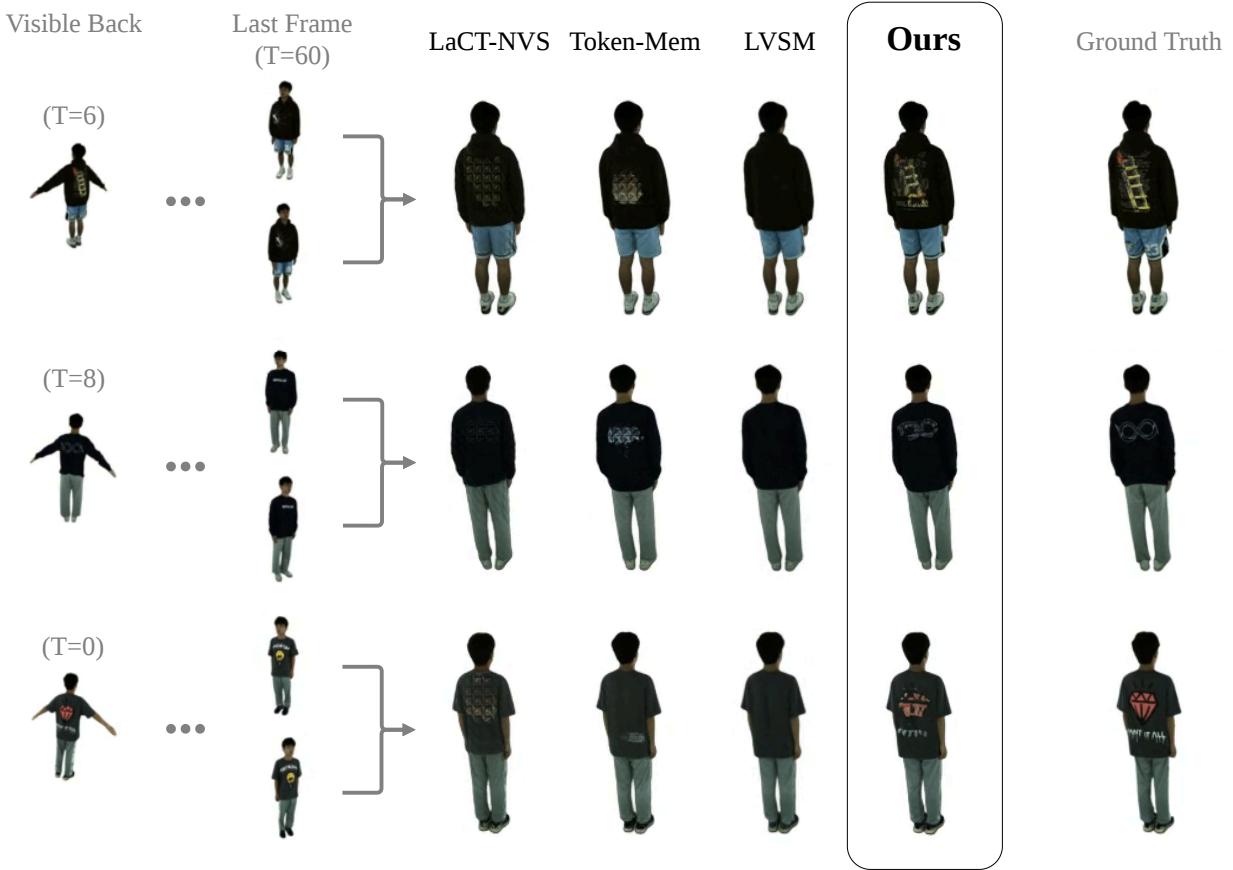


Figure 1. **Memory from Natural Rotations Qualitative Results.** Comparison with LVSM, Token-Mem (CUT3R), and LaCT-NVS on natural video sequences featuring dynamic reveals and occlusions. Left column shows at which timestep the back is revealed. NSTM accurately recalls T-shirt patterns and hair styles. LVSM hallucinates unseen regions, while LaCT-NVS and Token-Mem fail to recall back views. All stateful models are fine-tuned on 24 frames (see Sec. B.1)

above for LaCT and NTSM (ours). Architecture-wise, we follow the original CUT3R [3] implementation to use a state size of 768 tokens each with 768 feature channel. Differing from [3], we optimize L2 and LPIPS losses consistent with the other methods compared.

B. Additional Experiments

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B.1. Memory from Natural Rotations Test

To evaluate NSTM on the general setting of natural video sequences where occlusions can happen naturally at any point, we evaluate 17 scenes in MVHumanNet++ that feature natural rotations which show the back of the participants. This is in contrast to our “Memory Stress Test” our main paper where the model sees the back view only at timestep 0 and needs to remember it while receiving only frontal views over the entire duration of the video. To test, we provide fixed frontal stereo inputs (e.g., cam15 and cam00) and evaluate synthesis as the back naturally reveals and occludes.

Finetuning. For this test, we encourage longer memory persistence and enable the memorization of arbitrary, time-varying viewpoints. To this end, we extend the fine-tuning sequence from 4 frames to 24 frames and supervise the network at each timestep by a random, time-varying target view strictly withheld from the input stream. To construct the inputs, two random base cameras are selected and held constant for all $t > 0$. At the initial timestep ($t = 0$), we enforce a forced-occlusion scenario by exclusively sourcing inputs from two distinct, alternative cameras. Finally, to ensure the model properly learns object-centric rotation rather than just camera translation, 50% of the training sequences undergo coordinate alignment. Specifically, we modify the extrinsics of the first initial $t = 0$ camera to match the first base view. Throughout this extended sequence, we maintain the alternating memorization and synthesis steps utilized in the pre-training phase. We follow the same protocol to fine-tune the stateful Token-Mem and LaCT-NVS baselines.

Results and Discussions. We present our quantitative comparisons in Tab. 1 and qualitative visual evidence in Fig. 1 measuring performance both averaged across all frames and

Method	Average over T=60						Last frame					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	28.81	0.9654	0.0286	0.1216	19.21	0.5069	28.81	0.9656	0.0296	0.1298	19.69	0.5272
LaCT-NVS	29.13	0.9667	0.0253	0.1045	19.98	0.5850	27.01	0.9551	0.0340	0.1414	17.93	0.4982
Token-Mem	27.24	0.9565	0.0353	0.1303	18.05	0.4926	28.11	0.9615	0.0329	0.1363	19.14	0.5194
Ours	29.53	0.9692	0.0247	0.1056	20.39	0.5928	29.68	0.9691	0.0250	0.1107	20.88	0.5997

Table 1. **Memory from Natural Rotations Test Comparison.** best, second best, and third best are colored. NSTM outperforms Token-Mem and LaCT-NVS by 1.74 dB and 2.95 dB mPSNR on last frame measurement, while LVSM as a stateless model has close to identical average and last frame performance, demonstrating that it is not able to remember past details that would result in increased accuracy.

Method	Novel View Synthesis Over 60 Timesteps					
	PSNR↑	SSIM↑	LPIPS↓	DISTS↓	mPSNR↑	mSSIM↑
LVSM	29.48	0.9699	0.0225	0.0917	20.10	0.6509
LaCT-NVS	26.84	0.9567	0.0281	0.0997	17.62	0.5869
Token-Mem	22.81	0.9273	0.0477	0.1180	14.04	0.4595
Ours	29.11	0.9696	0.0223	0.0894	19.93	0.6703

Table 2. **General Novel View Synthesis Comparison.** best, second best, and third best are colored. We use two fixed frontal cameras as the input to NSTM, and use random target cameras at each time step to evaluate the novel view synthesis performance averaged over 60 timesteps across 360 scenes.

on the final frame in the ($t = 60$) sequence. The strength of NSTM remains evident, outperforming Token-Mem and LaCT-NVS by and LaCT-NVS by 1.74 dB and 2.95 dB mPSNR on last frame measurement. Since LVSM is a stateless model the average and last frame performance are identical, demonstrating that it is not able to remember past details that would allow it to increase its accuracy.

B.2. General Novel View Synthesis Test

While the main paper focuses on the extreme “Memory Stress Test” under forced occlusions, NSTM is fundamentally designed to be a highly versatile dynamic NVS framework. To evaluate its generalized synthesis capabilities, we conduct experiments under a standard, unconstrained streaming protocol. For this evaluation, we utilize the Stage 2 (Finetuned) checkpoints for all stateful models, as this stage specifically optimizes the network to handle time-varying target views without the rigid constraints of the initial memory bootstrapping phase (Sec. A).

Evaluation Protocol. To simulate typical real-world telepresence or broadcasting scenarios where input camera rigs are fixed, we restrict the streaming inputs to only two static frontal views (e.g., `cam15` and `cam01`). The model is then tasked with rendering the dynamic scene from completely random, time-varying target viewpoints. Because the target viewpoints are completely randomized at every frame, this unconstrained protocol inherently breaks the temporal continuity required for persistent memory accumulation. Note that this evaluation protocol is different from the training protocol, where the back view is always shown at the first

frame. We utilize this setup strictly as a stress test for per-frame spatial generalization.

Results and Discussions. Taking inputs from only the frontal cameras rarely add additional helpful temporal context. Despite this protocol lacking meaningful temporal context, Table 2 shows that NSTM consistently outperforms the stateful LaCT-NVS and Token-Mem baselines. Furthermore, NSTM remains competitive in standard structural metrics (LPIPS, SSIM, and mSSIM) against the purely stateless LVSM baseline, while outperforming on LPIPS and DISTS [1] demonstrating that NSTM’s conditioning on past observations drives higher perceptual-quality synthesis. This demonstrates our model’s spatial rendering capabilities.

C. Online Inference Protocol

Step	Latency (ms) ↓	FPS ↑
Memorization	58.14	17.19
Synthesis	27.01	37.02

Table 3. **Inference Performance and Scaling.** Inference performance breakdown measured on a single H100 GPU with a resolution of 256x256.

As shown in Fig. 2, NSTM explicitly decouples the frequencies of the Memorization Step and the Synthesis Step during online inference.

Decoupled Memorization and Synthesis Frequencies. Unlike models that rigidly couple memory updates and queries at every timestep, NSTM operates on independent execution strides. For our evaluation on the MVHumanNet++ benchmark, the sequences provided by the dataset are aggressively subsampled, capturing only 1 frame every 25 frames from the raw 30 FPS videos (i.e., subsampled at 1.2 FPS). Consequently, the temporal gap between consecutive inputs is ~ 833 ms. Because of this large temporal spacing, we utilize a 1:1 memorization-to-synthesis ratio for these specific benchmark sequences (i.e., memorizing every two frames). However, for processing real-world continuous streaming videos at 30 FPS, visual content is largely

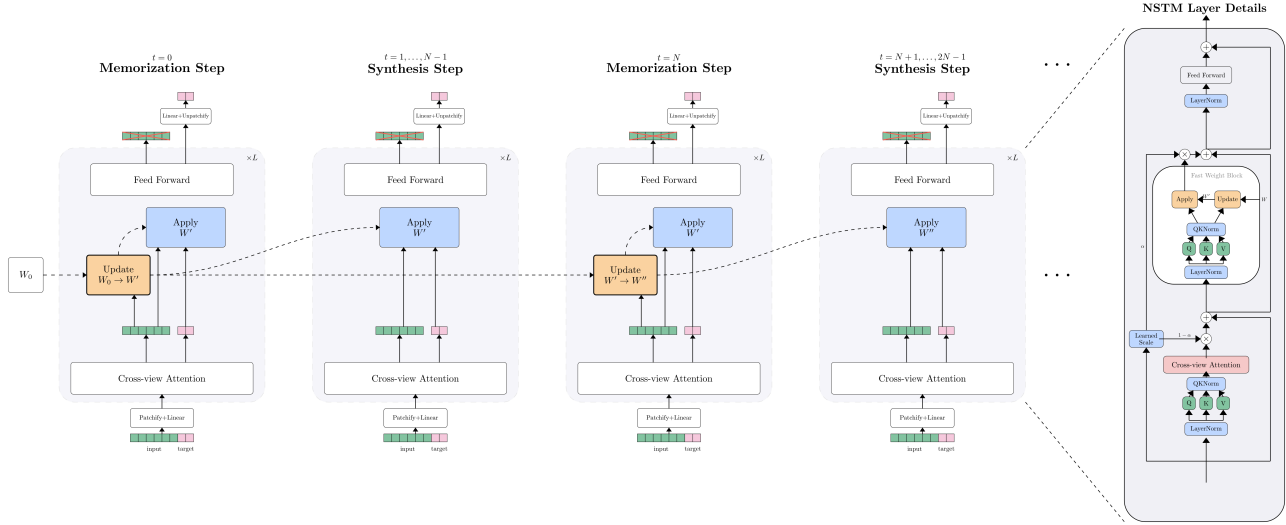


Figure 2. **Online Inference Protocol and NSTM Layer Details.** **Left:** During inference, we run the memorization step periodically while perform the synthesis step per-frame, which enables us to achieve amortized real-time speed. **Right:** We introduce the NSTM Layer consisting of cross-view attention, a fast-weight block and a feed-forward layer. We also incorporate a learned scale that allows the network to balance the contribution of memory and attention to each layer’s output.

redundant and there is no need to perform heavy memory updates for every frame. Instead, we successfully amortize computational cost by employing a 29:1 ratio: The memory module executes the Memorization Step only once every 30 frames (i.e., every 1 second) to capture novel states, and we query this persistent memory at 30 FPS to execute the Synthesis Step for the target views. Please see our supplementary website for more video results.

D. Detailed Model Architecture

Fig. 2 shows our NSTM layer in more details. Different from LaCT-NVS [4], we add an additional learned weight α to balance the contribution from the *cross-view attention* module and the *TTT* memory module.

E. Foreground Matting and Background Augmentation

To facilitate clean isolation of the dynamic subject from the studio background, we output a 4-channel $\text{RGB}\alpha$ image. During training, we employ background augmentation: we composite both the ground-truth foreground and our model’s predicted $\text{RGB}\alpha$ output onto randomly sampled solid color backgrounds before computing the photometric loss. This augmentation forces the network to learn a highly accurate alpha matte. Consequently, at inference time, NSTM produces a clean, disentangled alpha channel, enabling seamless compositing of the synthesized human subject into arbitrary novel environments.

References

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